

KRISHNA Koushik Thokala

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Summary

AWS Certified Solutions Architect - Associate (2026) with 2+ years leading 24/7 production incident response for 25-30 Tier-1 carrier clients. Built and validated AWS/EKS infrastructure using Terraform, GitOps delivery, observability, and scoped access controls, applying production operations discipline to cloud reliability.

Technical Skills

Cloud Platforms: AWS, VPC, IAM, EKS, ECR, ALB, Lambda, RDS, DynamoDB, S3, CloudWatch, Secrets Manager, KMS

Infrastructure & DevOps: Terraform, Kubernetes, Helm, Argo CD, Docker, GitHub Actions, GitOps, CI/CD, Linux, Bash

Security & Access: IRSA, OIDC, IAM Policies, Trust Policies, RBAC, NetworkPolicies, Security Groups

Observability & SRE: Prometheus, Grafana, CloudWatch Logs/Metrics/Alarms, k6, HPA, SLO/SLA, MTTR

Networking & Backend: TCP/IP, BGP/MPLS, DNS, Python, SQL, PostgreSQL, FastAPI

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Shift Lead (24/7 production NOC operations)

Pune, India

- Led 24/7 production NOC shift operations for 25-30 Tier-1 carrier clients, coordinating incident triage, escalation, field dispatch, TAC/vendor follow-up, and SLA-driven service restoration.
- Diagnosed high-severity BGP/MPLS, DWDM, and transport-network failures across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, loop testing, and segment-wise fault isolation.
- Led 5-person NOC shifts handling 40+ daily priority incidents/escalations, including structured handoffs, customer-premises testing, RCA notes, and cross-functional follow-up through resolution.
- Supported 99.9% annual network availability and 4-hour SLA restoration targets through incident command, vendor coordination, field-team execution, and RCA follow-up; earned Certificate of Excellence for carrier-grade reliability.

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

GitHub

- Built an EKS delivery platform where 4 Argo CD Applications reconciled Git-managed Helm configurations across isolated dev, staging, production-style, and observability environments with RBAC, NetworkPolicies, ResourceQuotas, and restricted pod settings.
- Provisioned and validated 20 AWS resources with Terraform; implemented immutable commit-SHA promotion and repository-scoped GitHub OIDC definitions for ECR; demonstrated drift correction and Git-revert rollback; and completed same-session teardown with an AWS Budget guardrail.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

GitHub

- Built an EKS operations platform connecting service ownership, deployment history, incident timelines, MTTR, and simplified error-budget views through FastAPI, React, PostgreSQL, and Alembic migrations.
- Deployed workloads behind ALB Ingress with Helm-managed health probes and resource limits, request correlation, structured logs, Secrets Manager-backed credentials, and security-group-restricted RDS access.
- Tested HPA behavior under k6 load as the backend scaled from 2 to 6 pods at 444% CPU against a 60% target, processing 2,035 requests with 99.5% success and 1.52s p95 latency.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

GitHub

- Built a serverless monitoring path where EventBridge invokes Lambda checks, DynamoDB stores current and historical status, SNS sends state-change alerts, and CloudFront serves a live React dashboard.
- Enforced read-only dashboard behavior with API Gateway throttling and daily quotas, restricted CORS, unsafe-target and redirect blocking, KMS encryption, DLQs, and a latest-status access pattern that avoids scanning retained history.
- Ran a controlled outage drill that detected failure in 1 second, displayed the DOWN state within 26 seconds, suppressed duplicate downtime alerts, and confirmed recovery notification.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India